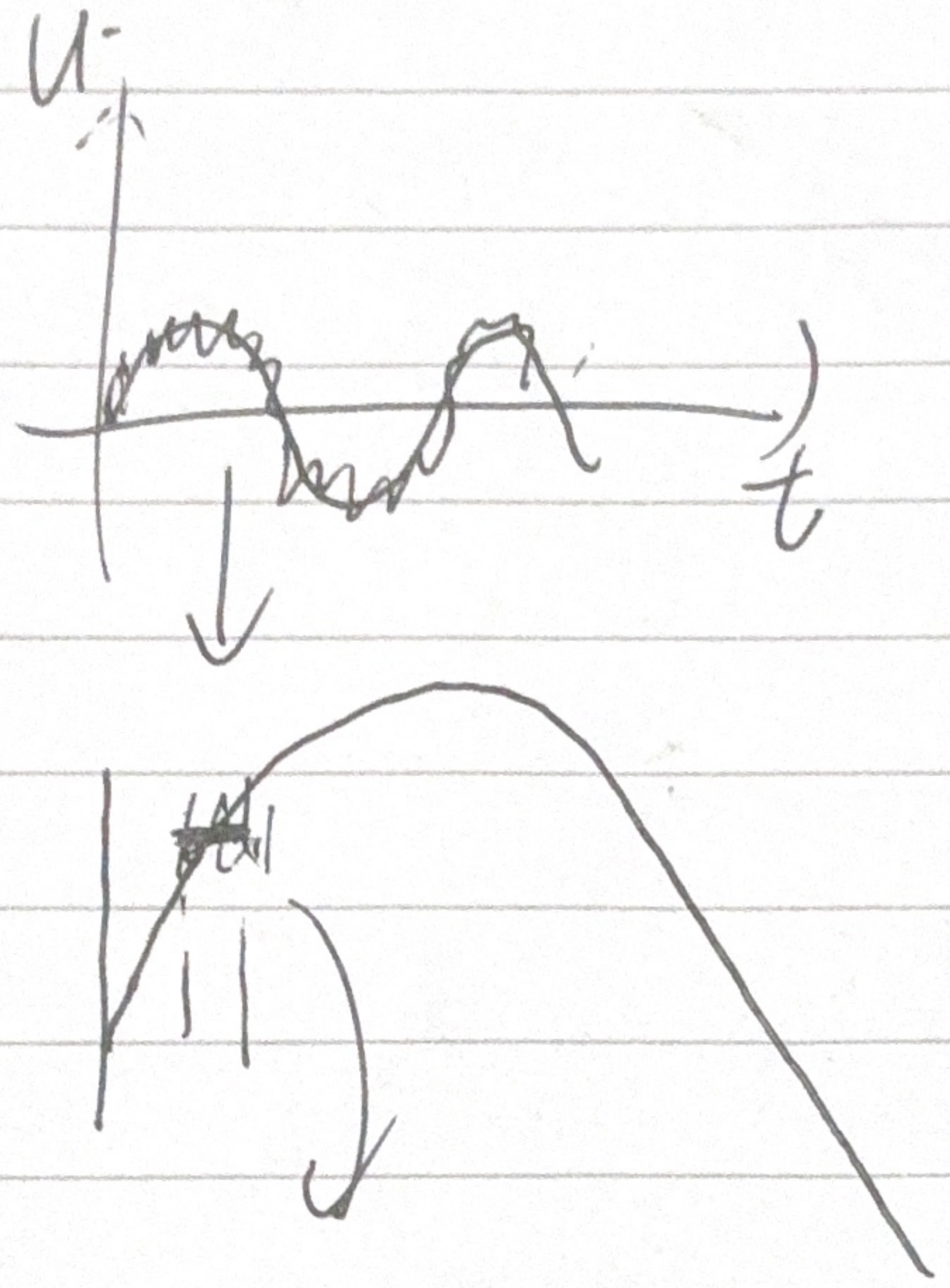
$\overline{U}(t)$



$$TKE \sim \frac{1}{2}(\underline{u'^2} + \underline{v'^2} + \underline{w'^2})$$

5min, 1min

$$\overline{u} = \overline{u}^{30min}(t)$$

0 – 30min

30 – 1h.

$$KE \sim \frac{1}{2}(u^2+v^2+w^2)$$

$$Re(\tilde{V}_0 e^{-i\omega t})$$

$$u \sim \cancel{\tilde{U}_0 \sin(\omega t)}$$

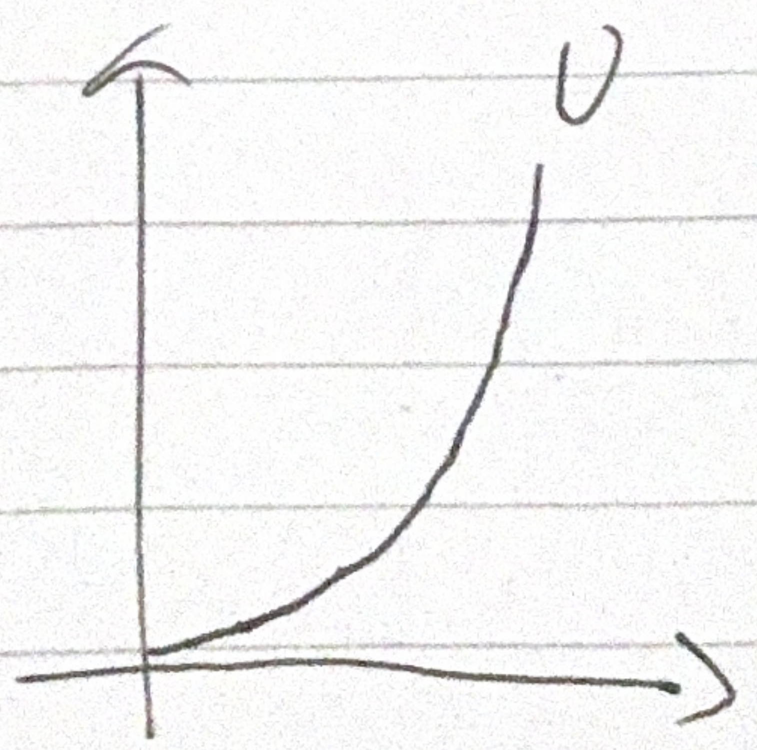
$$v \sim Re(\tilde{V}_0 e^{-i\omega t})\,\underline{x}$$

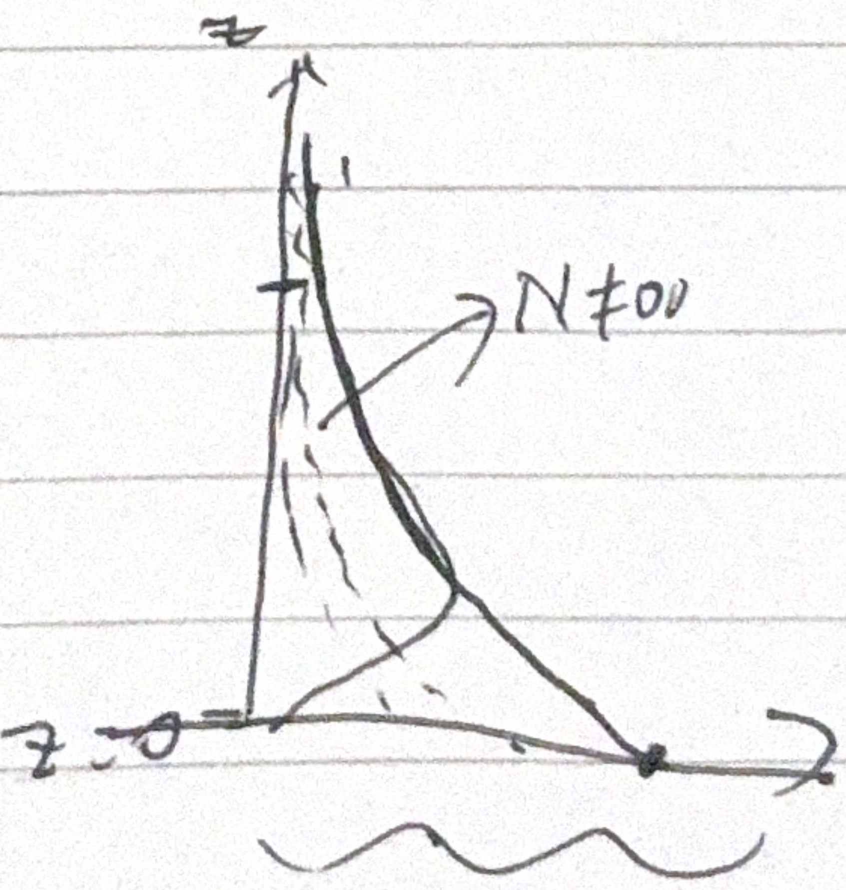
$$\overline{KE} \sim TKE + KE_{tide}$$

$$Ri = 0 \Rightarrow \boxed{\overline{TKE}(z)}$$

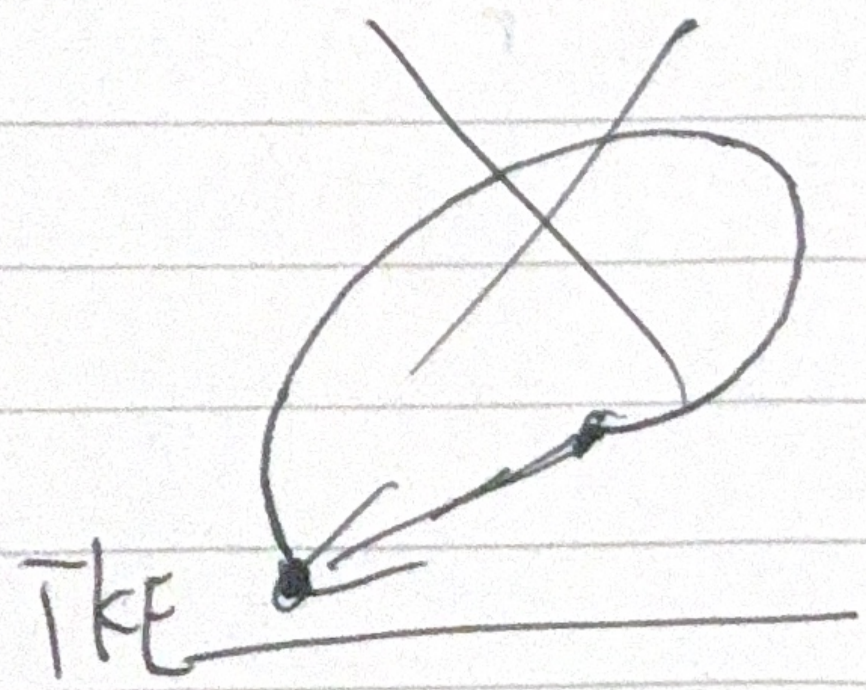
Ri

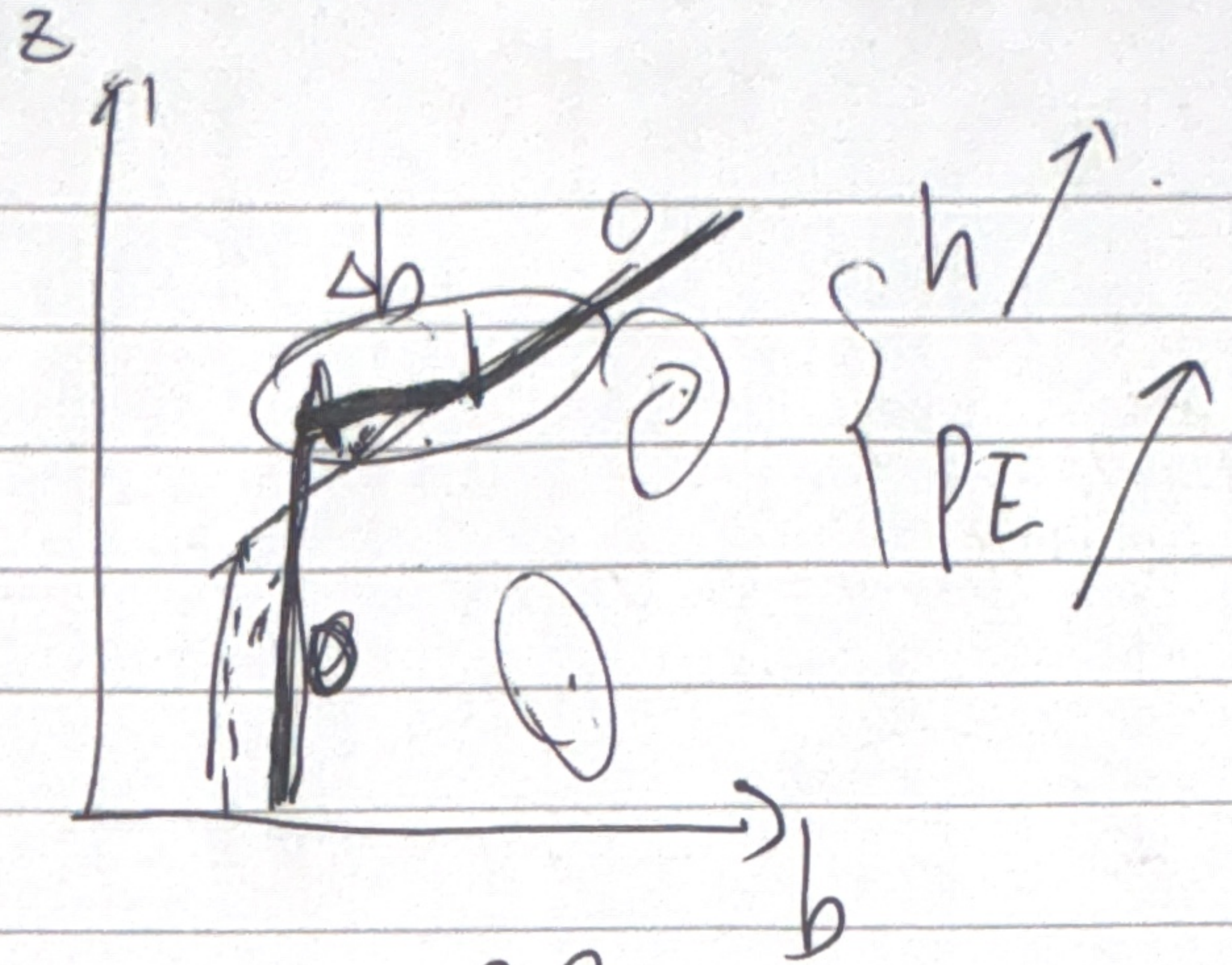
$\frac{N}{f}$ $\frac{N}{\omega}$



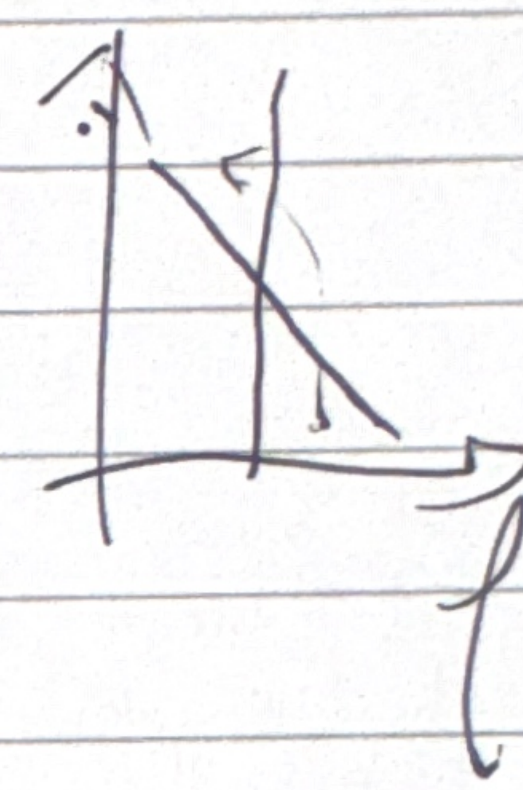


Ekman.



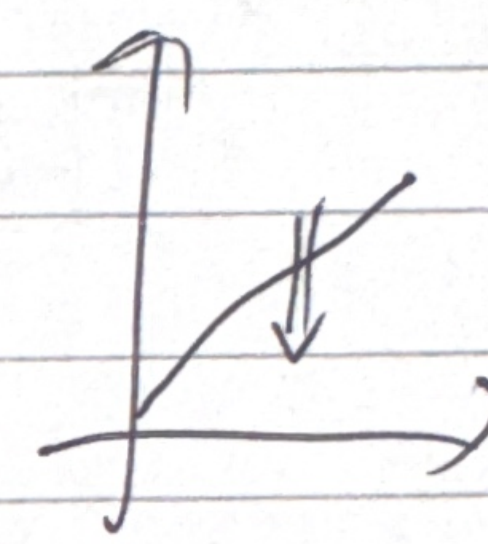


$$\left\{\begin{array}{l} PE = \langle g z \frac{\rho - \rho_0}{\rho_0} \rangle = -\langle z b \rangle \\ KE = \langle \frac{1}{2}(u^2 + v^2 + w^2) \rangle \end{array}\right.$$



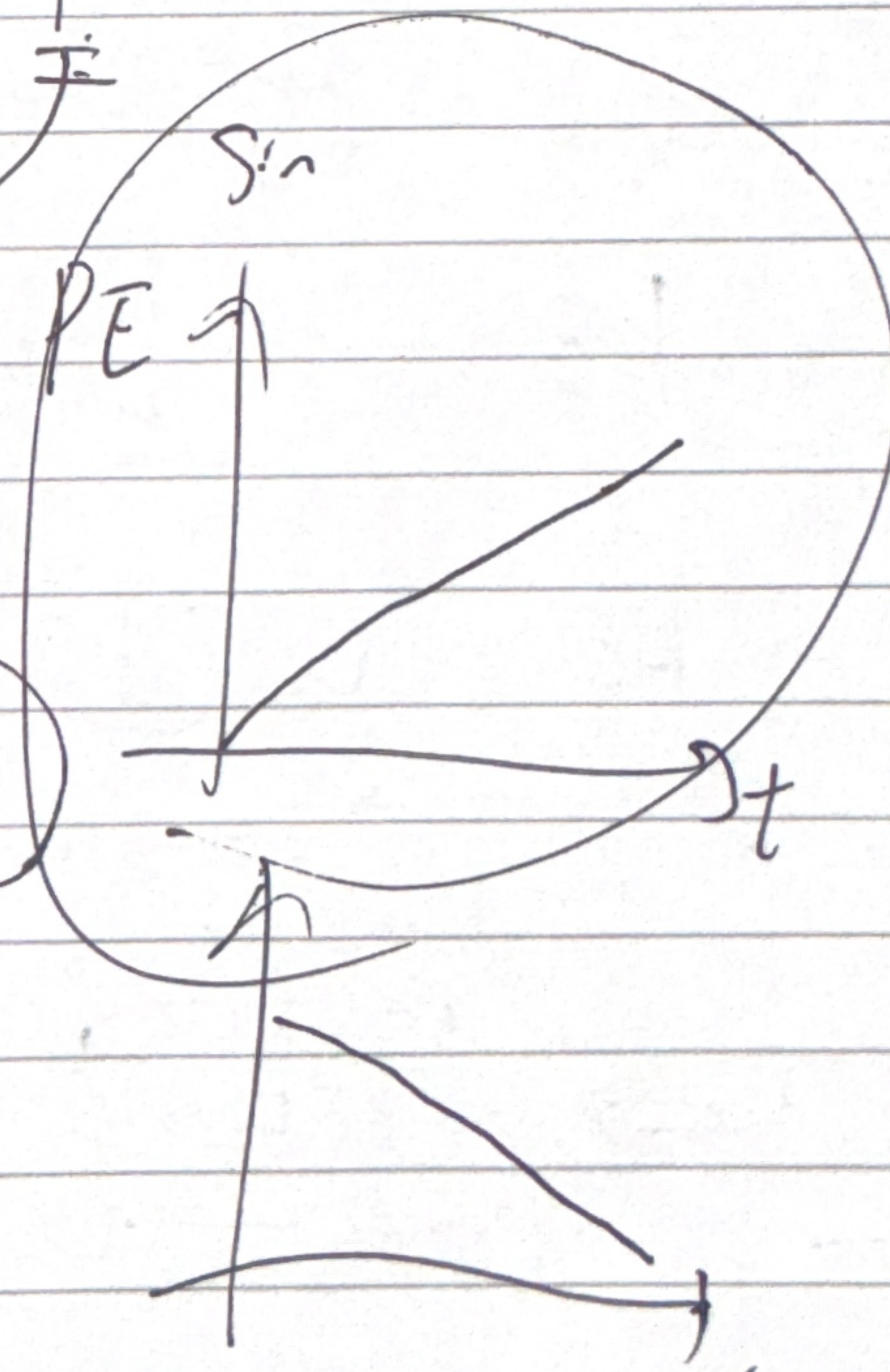
$$\frac{\partial}{\partial t} PE = \boxed{-\langle w'b' \rangle}$$

$$\frac{\partial}{\partial t} KE = +\langle w'b' \rangle - \varepsilon + P$$



$$KE = MKE + TKE$$

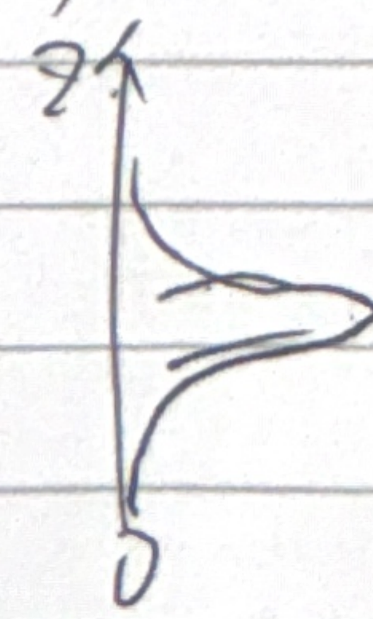
$$\frac{\partial}{\partial t} TKE = -\langle w'b' \rangle - \varepsilon$$



Predict: $\frac{\partial}{\partial t} PE$

$$① \quad \langle w'b' \rangle = \frac{1}{L_z} \int_0^z \boxed{w'b'} \, dz$$

$$= w'b' \big|_{peak}$$



①

effective diffusive

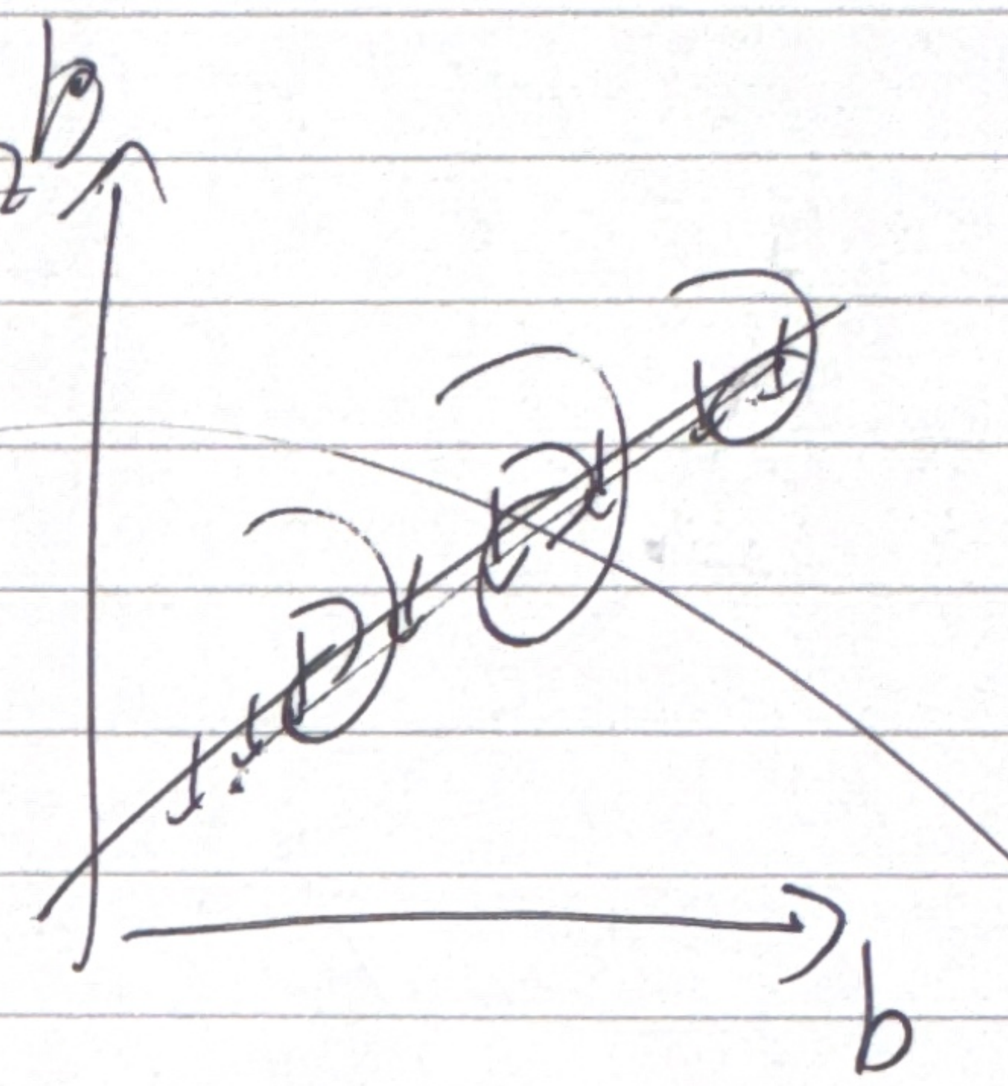
m^2/s $\quad m^2/s$

$k_T^{SGS} \frac{\partial b}{\partial z} + \overline{w'b'} = k_T \frac{\partial b}{\partial z}$

$k_T^{AMD} \frac{\partial b}{\partial z}$

k_T

$k_T^{turb} >> k_T$



$\frac{1}{L_z}\int_0^z w'b' \cdot dz$

$= \frac{1}{L_z}\int_0^z k_T \frac{\partial b}{\partial z} \cdot dz$

$\approx \frac{1}{L_z} k_T \Delta b$

k_T, $TKE(t)$, $z = z_1$

$h(t)$

$k_T (TKE)$

m/s

$k_T \sim U \cdot L \sim U \cdot U \cdot T \sim$

m^2/s

$\sim \frac{TKE}{N} \leftarrow R$

$\sqrt{TKE}$

m^2/s^2

$\frac{\partial U}{\partial z} \Rightarrow s^{-1}$

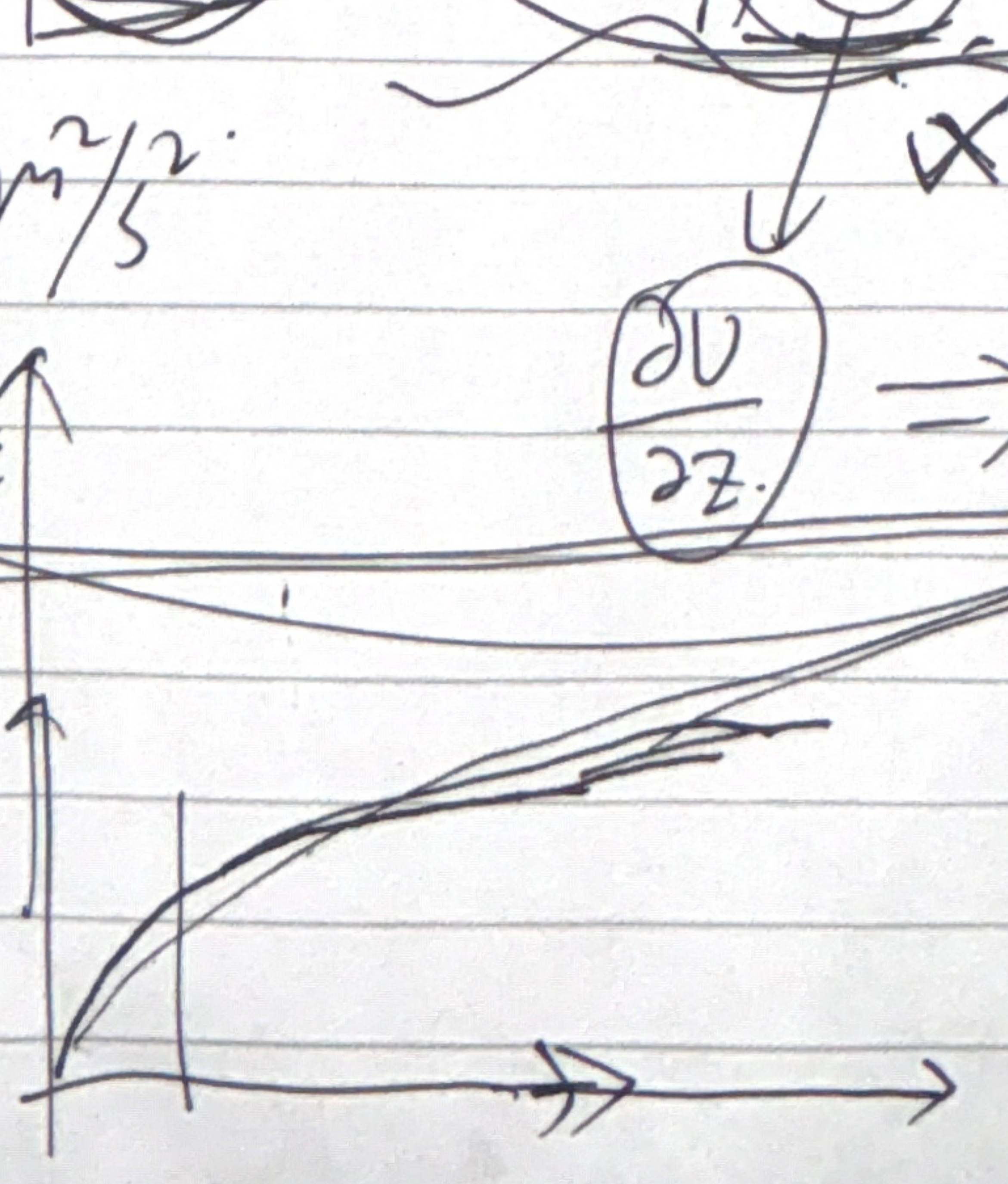
$N \sim \frac{1}{\rho_0} \frac{\rho - \rho_0}{\rho_0} g$

$\frac{\partial \rho}{\partial z} \cdot \frac{g}{\rho_0}$

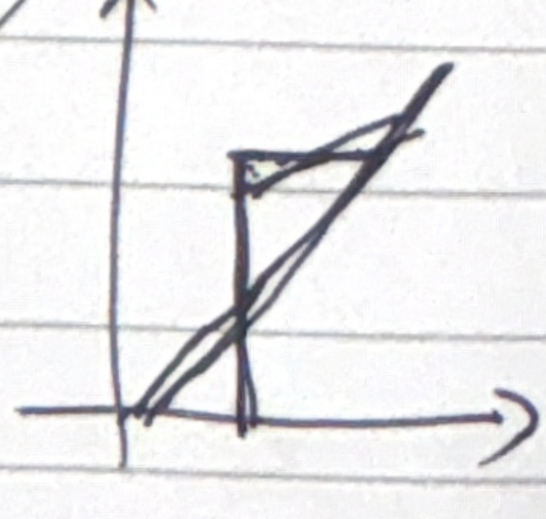
$[s^{-1}]$

$\frac{1}{N} \sim$

$h(t)$



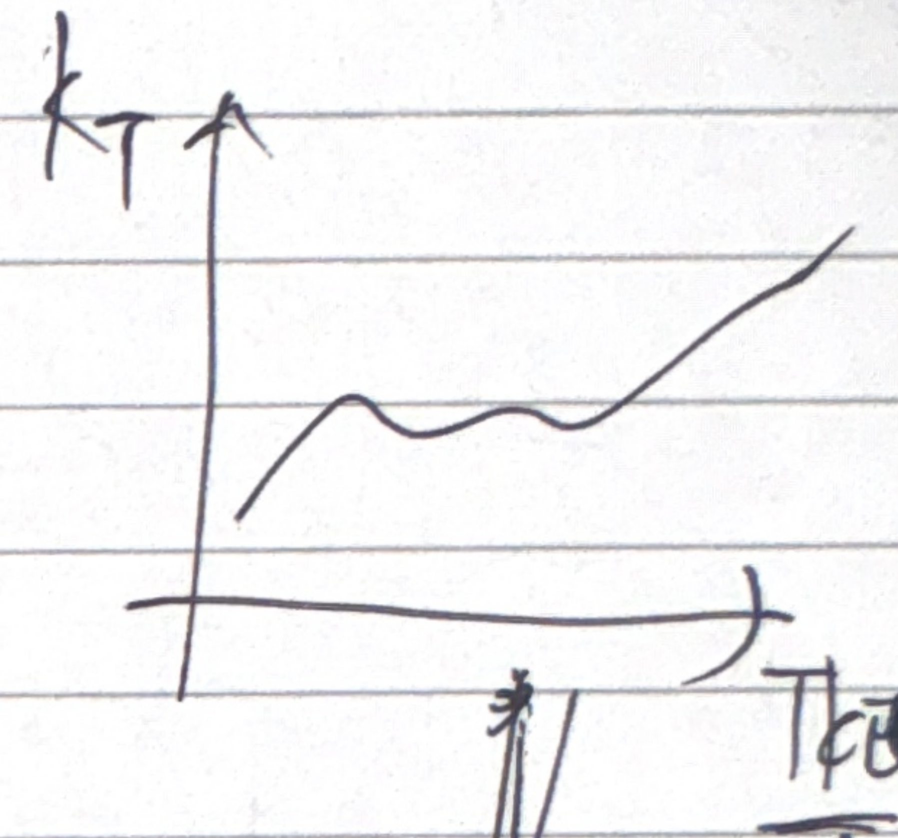
$t \cdot N$



$h \Rightarrow PE(h)$

$\boxed{k_T \ (TKE)}$

Simulation



theory

$\boxed{k_T \sim \sqrt{TKE}\ ?}$